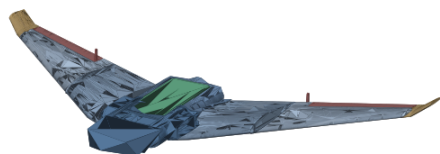


MB FLYING WING

Assembly & Print Guide · 688 mm span · 180 mm printer edition

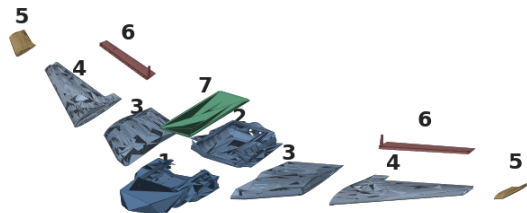


Wingspan	688 mm	Printed parts	11
Wing area	approx. 890 cm ²	Alignment pins	printed-in (d3.5 / d4.0)
Aspect ratio	5.3	Build volume needed	180 x 180 x 180 mm
Mean aero. chord (MAC)	approx. 158 mm	Layout	pusher flying wing, elevons
Target AUW	400 - 600 g	Structure	hollow shell + internal lattice

This guide covers the 180 mm printer edition of the model: oversized parts are pre-split with printed-in alignment pins, the fuselage carries a new streamlined nose cone, and the cabin hatch is engraved with the MB monogram. All joints are designed to be glued - the pins align the parts, the adhesive carries the load.

1 · Parts list

#	Part	File	Qty	Notes
1	Fuselage nose half	Fuselage_Burun.stl	1	includes aero nose cone + 2 pins + MB floor emblem
2	Fuselage rear half	Fuselage_Arka.stl	1	motor mount at rear, 2 pin sockets
3	Wing root section	Wing_1_Kok / Wing_2_Kok	1 + 1	3 alignment pins on joint rib
4	Wing tip section	Wing_1_Uc / Wing_2_Uc	1 + 1	3 pin sockets on joint rib
5	Winglet	Tip / Tip2	1 + 1	glues to wing tip
6	Elevon	Elevon / Elevon2	1 + 1	control surface, hinge at wing trailing edge
7	Top hatch	Top_Hatch.stl	1	MB engraved on the inner face



Exploded view - numbers match the parts list. Left/right parts are mirrored pairs.

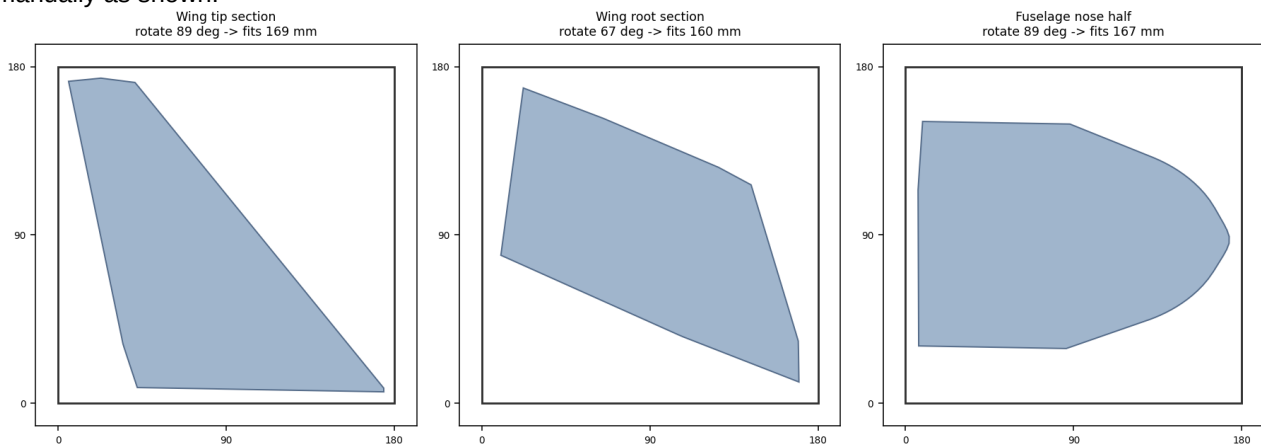
2 · Printing

Setting	Value	Why
Material	LW-PLA or PLA	LW-PLA saves ~40% weight if available
Layer height	0.15 - 0.20 mm	clean aero surfaces
Perimeters	1 - 2	shells are the structure
Infill	0 %	parts are already hollow with internal lattice
Top / bottom layers	0 (wings) / 2-3 (fuselage, hatch)	wings close themselves via the skin
Supports	OFF	all parts print support-free in the supplied orientation
Brim	4 - 6 mm on wing sections	thin trailing edges grip the bed better
Seam	rear / aligned	keep it off the leading edge

Slicer settings above are a starting point for a typical 0.4 mm nozzle - adapt to your printer and material. If your slicer complains about the ultra-thin walls, enable "detect thin walls" / "print thin walls".

Fitting the 180 x 180 mm bed

Three parts only fit when rotated diagonally on the bed. Your slicer's auto-arrange may not find this - rotate manually as shown:



Print orientation: wings and elevons flat as supplied (largest face down), fuselage halves flat on their underside, winglets and hatch flat. Do not scale any part - pin diameters and joint clearances (0.25 mm radial) are tuned for 100 %.

3 · Assembly

Adhesives: medium CA (cyanoacrylate) for quick tacking, 5-30 min epoxy for the load-bearing wing and fuselage joints. Dry-fit every joint first.

Step 1 - Fuselage. Glue the nose half (1) to the rear half (2). Two d3.5 mm pins on the nose half enter the sockets of the rear half; apply epoxy to the U-shaped joining rib faces, press together and check the halves sit flush all around. The interior bay stays open for battery and electronics.

Step 2 - Wings. Join each wing root section (3) to its tip section (4). Three pins per wing align the full-profile joint ribs. Epoxy the rib faces, clamp lightly with tape, keep the wing flat on a table until cured so no twist is built in.

Step 3 - Winglets. Glue winglet (5) onto each wing tip. Match the angled mating faces; CA is fine here.

Step 4 - Wings to fuselage. Slide each completed wing onto the fuselage side rails and epoxy. Check alignment from the front: both wings level, winglets symmetric.

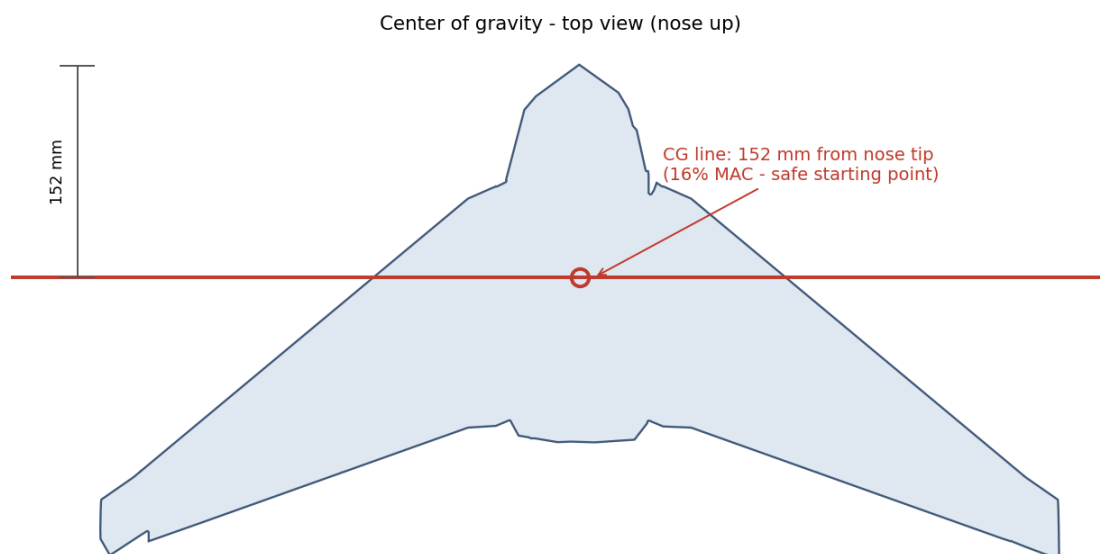
Step 5 - Elevons. Hinge each elevon (6) to the wing trailing-edge cutout with tape hinges (top-side full-length tape, bottom gap tape) or CA hinges. Aim for at least 20 degrees of travel each way.

Step 6 - Hatch. Drop the top hatch (7) into the fuselage opening - MB engraving faces the inside. Retain with a small magnet pair or a strip of tape; do not glue, this is your access to the bay.



Joint faces: pins (blue faces) enter sockets (green faces). Sockets have 0.25 mm radial clearance - if a pin binds, a light twist with sandpaper is enough. The pins align; the glued rib area carries flight loads.

4 · Balance and first flight



Center of gravity

Start with the CG **152 mm behind the nose tip** on the centerline (about 16 % of the mean aerodynamic chord - a deliberately nose-heavy, safe starting point for a flying wing). Balance with the battery position first, add nose weight inside the hollow nose cone only if needed. After trimmed test glides you can move it back a few mm at a time for better efficiency - never fly a wing tail-heavy.

Setup

Elevon mixing (delta mix) on the transmitter, 30-40 % expo. First flights: 3-5 mm reflex (both elevons slightly up). Typical power for this size: a 2205-2306 class motor on 3S with a 6x4 pusher prop mounted on the rear mount - verify thrust and clearance for your own components. Hand-launch level and firm, into the wind.

Wind tunnel companion

The interactive wind tunnel ([Ruzgar_Tuneli_MB.html](#)) uses this exact airframe geometry. Use it to sanity-check your all-up weight: enter the finished weight and it reports the minimum flying speed and stall margin before you ever throw the model.

Safety: this is an experimental 3D-printed aircraft. Inspect every glue joint before each flight, fly in open areas away from people, and follow your local RC regulations.